

Sprint 5 Documentation

Sprint Number: Sprint 5

Sprint Duration: 2 Weeks

Sprint Focus: External Lesson Content and Backend Refactoring

1. Sprint Objective

The objective of Sprint 5 was to improve the maintainability of lesson content by moving hardcoded lesson data into external CSV files. This sprint focused on content organization, API routes, and lesson system refactoring.

2. Sprint Scope

The scope of this sprint included creating a structured lessons directory, organizing lesson content by language and lesson, building backend API routes to serve lesson data, and refactoring the lesson page component to load content dynamically.

3. Functional Requirements

Lesson content should not be hardcoded inside TypeScript arrays.

Lesson content should be stored in external CSV files.

Each supported language should have an organized lesson folder.

Each lesson should contain structured content sections.

The frontend should load lesson content through an API.

The shared lesson page should display dynamic lesson data.

The system should be easier to update without changing source code.

4. System Design Activities

The team designed a new lesson content architecture. Instead of storing all lesson content inside the frontend code, the content would be placed inside organized CSV files. Each language would have its own folder, and each lesson would have its own content files.

The backend API design was also planned. API routes would read the CSV lesson content and send it to the frontend. This separated content management from the user interface and made the system more scalable.

5. Implementation Work Completed

A new lesson content structure was created.

Lesson content was moved into CSV files.

Language folders were organized.

Lesson folders were organized.

API routes were added to serve lesson content.

The shared lesson page component was refactored.

The frontend was updated to load dynamic content.

Turkish and Persian lesson structures were prepared.

6. Testing Activities

The team tested whether lesson content loaded correctly from the new CSV structure. API routes were tested to make sure they returned the correct lesson data. Lesson pages were tested to verify that vocabulary, key sentences, dialogue, listening, speaking, and quiz sections displayed correctly.

7. Challenges

One challenge was restructuring the lesson content without breaking the existing lesson pages. The team had to make sure the frontend could still display content correctly after the data source changed.

Another challenge was organizing the CSV files in a way that would support multiple languages and lessons in the future.

8. Sprint Deliverables

External CSV lesson content files.

Organized lesson directory.

Backend API routes for lesson content.

Refactored shared lesson page.

Dynamic lesson content loading.

Improved maintainability.

Reduced hardcoded lesson data.

9. Sprint Review

Sprint 5 successfully improved the content management system of the application. Lesson data became easier to update, and the codebase became cleaner because large hardcoded arrays were removed from the frontend.

10. Sprint Retrospective

Sprint 5 showed the importance of separating content from code. The team learned that external content files make the project easier to maintain, especially when the application supports many languages and many lessons.